

European Nations Have Solved Problem of Salving Submarines

Improved "Mother Ships"
Have Raised Disabled
Vessel from Bottom to
Surface in Three Hours.

EDISON PERFECTS HIS
AIR PURIFYING DEVICE

By A. M. JUNGMAHN.

THE recent disaster to the F-4 calls to mind the difficulties experienced by the navies of the world in raising wrecked submarines. Up to the year 1908, when the German salvage dock for submarines was built, the work of raising a wrecked submarine was not only very difficult, but exceedingly slow. The old practice was to submerge floats and pass chains beneath the hull of the sunken vessel. The vessel was raised when the water was pumped out of the compartments of the floats. The whole affair—floats and submarine—then had to be hauled to a drydock, the submarine blocked and the floats removed before an examination of the wreck could be made.

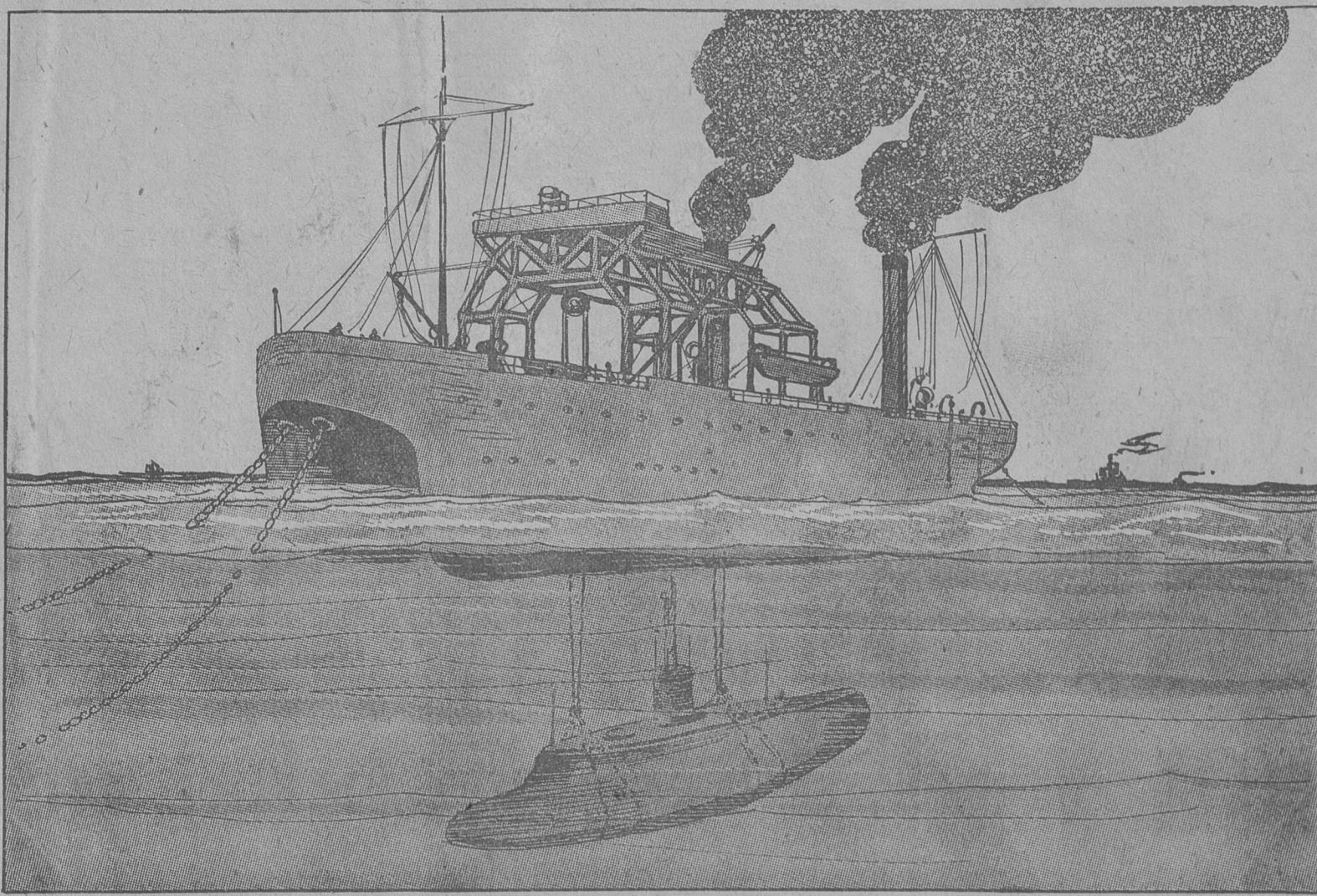
The first step in simplifying the salvage of wrecked submarines was to fit the vessels with rings to which chains could be attached, as the work of passing chains under the hull was found to be laborious in the extreme. Then came the salvage dock Oberelbe, constructed for the German navy. Shortly afterward a similar dock was put into use in the French navy. The German dock was 230 feet in length, with 600 tons capacity. The French dock was a little larger and much more powerful, being 262 feet in length and capable of lifting 900 tons.

These two first salvage docks were constructed with a central floor and provided with compartments on either side, which were to be filled when the dock was to be submerged or emptied when it was to be floated. In the Oberelbe the compartments are connected firmly by crossbeams and the floor is removable. The dock is equipped with powerful cranes and tackle. The electricity for operating the cranes is generated in one of the compartments.

The Method Used.

When the dock is to raise a wrecked submarine it is towed to the damaged vessel, the floor of the dock is removed, and the dock itself is about two-thirds submerged. The submarine is made fast to the tackle by means of the rings provided for that purpose, and is raised by the cranes. While the cranes are lifting the vessel the compartments of the dock are emptied. When the entire submarine has been brought to the surface the floor of the dock is slipped beneath the hull and the dock is pumped dry.

Since the construction of the Oberelbe ad-



How the "Mother Ships" in Use in European Navies Raise Disabled Submarines.

vancement has been made in the building of submarine salvage docks. It has been the endeavor of the German navy to have such a dock at each submarine base, so that in case of accident the crew of a wrecked submarine would have the chances of rescue in their favor.

Rescue in Three Hours.

The excellent work by the improved type of salvage dock was strikingly exemplified by the raising of the German submarine U-3, which was sunk in the harbor at Kiel January 17, 1911. In this instance the salvage dock Vulcan, a seagoing vessel capable of navigating under its own steam, raised the U-3 in less than three hours after she was moored above the crippled submarine.

The U-3 was built at Danzig in 1909, and attained distinction by making the trip from Cuxhaven to Kiel, a distance of 540 miles, in forty hours. She was accompanied by the submarine U-4, and had no convoy. The U-3 is a vessel of 300 tons displacement.

At the time of the accident the Vulcan was anchored at Kiel and two submarines were practicing in the harbor. One of the vessels always remained afloat while the other was under water.

The distress of the U-3 was immediately observed by her companion submarine, the U-1. The Vulcan, known as the "mother" ship, was not immediately employed to raise the U-3, because the

submarine was not, at first, entirely submerged. The prow and periscope emerged from the water, while the stern was held fast in the mud bottom of the harbor. An attempt was made to rescue the imprisoned crew by hauling the bow of the U-3 out of the water by means of a floating crane, which was attached to the sunken ship by chains passed around the hull by divers. This method was partly successful, and the vessel was held up long enough to allow the crew to be drawn out of the torpedo compartment. Three officers elected to remain in the middle and turret compartment in order to empty the forward tank by means of compressed air, thus allowing the vessel to rise until the bow was twenty inches above water so that the twenty-eight men of the crew were saved. They lost their lives through that act of heroism.

Crane Is Disabled.

After the crew was rescued an attempt was made to raise the vessel still further by means of the floating crane, but the breaking of the hawsers and the disabling of a crane prevented this, and the U-3 sank back and was entirely submerged. The Vulcan, which was lying by, was now called into action.

Because the water was so shallow it was a very delicate operation to place the Vulcan in position over the U-3 without danger of breaking

the periscope, which could be seen beneath the surface. If the periscope had been broken the turret and central compartment of the unfortunate submarine would have been flooded with water and the three officers, whose fate was not then known, surely killed.

To add to the difficulties of the Vulcan's work darkness had fallen, and the salvage had to be carried on with searchlights. Also a strong wind was blowing, which seriously interfered with floating the Vulcan directly over the sunken vessel with her twin hulls spanning it.

Finally, with the aid of anchors and several steamers, which acted as tugs, the Vulcan was brought into position with her hoisting tackle directly over the lifting hooks of the U-3. It required about two hours for the tackle to be attached to the hooks by divers, and then the actual work of lifting was begun. In exactly thirty-five minutes the turret rose above the water, and in ten minutes more the fate of the three officers was ascertained.

The Vulcan is 280 feet in length, and is, of course, double-hulled. She has a lifting capacity of 600 tons. The vessel is so constructed that there is sufficient width between the two hulls to admit the submarines. Along each wall of the interior opening a shelf is formed upon which the removable floor of the dock rests. Like any other floating dock, the two hulls are constructed

with watertight compartments. Two powerful gantry cranes span the well between the hulls. The cranes are fitted with heavy lifting tackle, and, as in the case of the first salvage dock, the Oberelbe, they are operated by electricity. A generating set in one of the compartments supplies the current. The advantage of the Vulcan type of salvage dock over the earlier ones is that it navigates under its own steam, and does not have to be towed to the scene of the wreck.

Great Pressure a Handicap.

In the case of a submarine which lies deep in the water, it is especially difficult to attach hoisting lines, because of the great pressure of the water which the divers have to endure. The rate of increase of the pressure of the water is 0.435 pound per square inch each vertical foot. For example, at 300 feet below the surface the pressure would be 108 1/2 pounds per square inch.

In order to facilitate the training of deep-water divers a pressure tank has been constructed which has been in use in the Brooklyn Navy Yard. The tank is a steel cylinder so arranged that the water it contains may be placed under any desired pressure up to what would naturally be encountered at an ocean depth of 250 feet or more. Although this tank is but seven feet in diameter and contains but eight feet of water, the diver in training experiences all the sensations which he would experience in actual ocean diving, and avoids much of the risk. For purposes of observation from the outside, the tank is fitted with glass windows and is lighted on the inside by electricity.

The type of equipment now in use in the United States Navy is practically the same as that employed in the British navy. For deep diving such as that necessary to raise the F-4 at Honolulu, it is thought that the air supplied the divers will be furnished by a power compressor instead of by the usual hand pump.

Edison Improves Equipment.

In 1910 Mr. Edison became interested in the problem of constructing batteries for submarines which would eliminate the sulphuric acid gas which is the cause of inconvenience and suffering on all submarines. Last summer the United States submarine E-2 was submerged near Newport, and at that time the crew suffered severely from the effects of the poisonous chlorine gas. This vessel now is being fitted with the new Edison batteries. With the new Edison equipment there will be a sufficient amount of potash solution on board to purify the air for 100 days. The batteries contain a potash solution, and the only metals used are steel and nickel. Potash absorbs carbonic acid gas. Pumps will drive the gas-laden air, which descends to the bottom of the boat, through the cells filled with potash solution, thus effectually purifying it.

At Portsmouth, N. H., the United States Government is now building a submarine which will be the finest vessel of its type in the world. It will be known as the L-8, and is to be fitted with the Edison apparatus. Mr. Edison has devoted much time and energy toward perfecting these batteries from humanitarian motives, and it is his desire to see all the navies of the world adopt the apparatus for their submarines. No restrictions will be placed upon the invention.

German "Knapsack Dictionary" Tells Soldiers English Words to Use in Britain

Everything from How to
Order Lunch or Demand
Lodgings to the Phrase
'Give Me a Kiss' Is Listed.

ENGLAND was amused and a little bit shocked recently when an army chaplain, the Rev. G. Anderson, turned over to the newspapers a "German Knapsack Dictionary" containing words and phrases likely to be useful to the Kaiser's troops should they land in Great Britain.

The dictionary, so goes the story, was the property of a German corporal wounded near La Basse. He gave it to the chaplain as a souvenir.

The book, it seems, begins with a list of names of most of the important localities in England. Then follow two maps, one of the British Isles and another of London. Seven pages of description of titles and rank of British officers follow.

Of the 4,000 or more words in the actual dictionary, more than half of them apply solely to naval and military operations; and there are two chapters of phrases for invading troops.

A key to English pronunciation and grammar precedes the dictionary. One is immediately impressed with the large proportion of words that would have to be known to a visionary German army of occupation in Britain. For instance, "unsaddle, repel an attack, cut off the retreat, shoulder strap, aide de camp, aeroplane, set on fire, and take aim at." "Puttees" are translated "Wickel-gamaschen."

Every German synonym for that sinister word "requisition" appears, as well as the words "booty" and "plunder."

Evidently the producer of the book knew a great deal more about modern military and naval developments than is vouchsafed to the ordinary lexiconer.

A Gallant Phrase.

"Geisch" (hostage) is a hint that a German army would operate in England in the same way that it has behaved in the territory it has invaded on the Continent, i. e., by taking hostages.

The Germans are told the English for "hand-to-hand fighting" and "extended formation."

Even the dictates of gallantry are not forgotten, for the phrase "Give me a kiss" is set out especially for their benefit.

In any illustration of the use of a preposition of time or place London is always chosen for the example, e. g., "London liegt an der Themse." "Mord" means "murder" and "Mordbrenner" means "incendiary," words the Germans might think about. The pseudo invaders of England are instructed in the phrases to "cry quarter" and "no quarter" and "flag of truce."

Next comes "Gesprache—Das Quartiermachen," i. e., billeting. What is not in the interrogative is in the imperative, as will be seen from the following representative extracts taken textually from the little red book, each sentence being followed by the phonetical spelling:

Where does the Mayor (head local authority) live? Show me to him! Send him here!

Give me billets for 5 officers, 220 men and 8 horses.

Send somebody with me who will show me the rooms for the officers.

Is that the room for the officer? Don't you have a larger room? The rooms must be heated to-morrow. Put clean sheets on the bed.

Lunch for the Captain.

See that the captain may have lunch immediately after arriving.

Have any sick horses been in this stable? Take that cow away!

Give notice by the drum that the inhabitants will have to furnish the food for the soldiers quartered with them.

Every soldier will be entitled to one and a quarter pounds of meat—in addition to that a pint of beer. For the night every soldier must have at least a straw bed, a pillow and a blanket.

Take care that my orders are strictly obeyed. I shall hold you responsible for that.

Any resistance will be severely punished.

The community (order municipality) will have to pay a heavy contribution.

Whoever makes an attempt on one of my soldiers will be condemned to death.

The village will be razed.

The Mayor will certainly be shot.

You will answer with your life for their safety.

Thus, having shown terror-stricken Englishmen something of the rigors of a German occupation, the manual tells the triumphant soldier what to say after his billet has been indicated to him.

Show me to my room! That dirty hole? What do you think?

What? You have no other room in this big house? Open the doors! I am going to choose a room myself. Here I shall stay. Take these things out of the room!

Quick! I should like to have something to eat. A light refreshment will do for the present. Bread, butter, cheese and some meat. Some wine, a bottle of beer, a glass of milk. I am very thirsty.

See that I am not disturbed! Don't make any noise! Now I want to sleep for an hour. At 2 o'clock you will have dinner ready. Call me when dinner is ready.

Then comes praise indeed and a touch of politeness:

The food is well cooked. May I ask for a little more beer? Don't you have a better sort of beer?

Next a strange request for an English housewife:

Will you give us some matches and a boot-jack?

To-morrow morning we must have breakfast in due time. We should like to have coffee or a milk soup.

Make some sandwiches for us to take along with! Fill this bottle with coffee!

After a section dealing with stabling of horses comes the section "Auf Vorposten und Patronille"—on outpost and patrol duty. Several forms of challenge lead up to the following:

Halloo! What are you roving about here? Come here! You will come with me to the piquet.

Demands to See Purse.

Walk in front of me! Be silent! You have only to answer my questions!

You seem suspicious to me. Show me your pocketbook!

You will stay here until further notice. If

you make any attempt to escape you will be shot.

A series of queries with regard to roads and distances, and then imagine a dumfounded East Anglian peasant, with bayonets in unpleasant proximity to his ribs, being obliged to submit to the following catechism:

Is that village occupied by troops? When did the troops arrive there? What number of men do you think are there? Well? About two or three companies? How many officers were with them? Was there any artillery? How many guns? Did you see any cavalry?

Speak the truth! A lie could cost your life.

Has the village been prepared for defense? Are there any barricades at the entrances? Is there a church in the village? Is there a castle or any strongly built house?

Where are the enemy's outposts? Didn't you see any hostile patrols in that part?

Military Information.

How strong was the patrol? Where did it come from and in what direction did it ride on? Was it commanded by an officer?

Keep close by the side of my horse! On the first attempt to escape, or if you take me the wrong way, I shall shoot you.

So the unjoyful peasant will know what to expect if he does not guide the invaders to his fellow countrymen's camp:

Stay here! I will call the miller myself. Halloo, miller! Have any hostile troops marched through this place?

You do not speak the truth! Here are distinct and quite new traces. I am sure there must have been a bivouac here.

The Sherlock Holmesian Deutscher does not proceed to order the miller's instant execution, but plies him with further questions. Then apparently the reconnoitering party ride on and take a British dispatch rider by surprise:

Halt! Surrender! Don't move or you will be shot at! Dismount and put down your arms! What is your mission? Give me your message, your letter! You are a prisoner! Mount again, and give the reins of your horse to that hussar!

At last comes the chapter on "Betreibung"—requisitioning. Subjoined are salient passages:

Mr. Mayor (oder sir), I have orders to

carry out a requisition of food and forage in this place. I hope you will not compel me to apply force.

Within an hour you will have to furnish on the spot — head of cattle, — hundredweights of oats, flour, hay, — hectors of wine, potatoes.

In case of refusal you will be arrested. I shall cause my soldiers to take the articles required where I shall find them. You see, your refusal would be of no use. Any resistance of the inhabitants would be severely punished.

You will get a receipt for the articles furnished. By virtue of this receipt you will be compensated later on.

Take a resolution! I have not much time! You will not be able to call for assistance. All exits are watched.

You have a lot of fowls; you will furnish 300 eggs! Your fowls don't lay? Well, then, we will take all these useless creatures with us and slaughter them. Eggs or fowls! You may choose!

So much for German requisitioning in Albion. Next chapter is "Gefangenentransport"—transport of prisoners of war:

Death for Recalcitrants.

Silence! Nobody is allowed to talk! Away with the pipes (cigars)! Take your hands out of your pockets!

When I speak to you you have to assume a military attitude.

Every insubordination and every breach of discipline will be severely punished.

In the restaurant military severity is relaxed. The German invader asks:

Which is the best hotel in this place?

He goes there, and orders a substantial meal with a bottle of claret. He is a trifle impatient, however, and cries:

Waiter, make haste! We don't like to wait a quarter of an hour for every glass of beer.

We want another waiter. Otherwise we shall send for some soldiers to attend on us.

Then the refreshing passage:

I want to pay (oder, my bill, please). Supper, one brandy (whisky) and three glasses of beer. Here are five shillings (is a crown)!